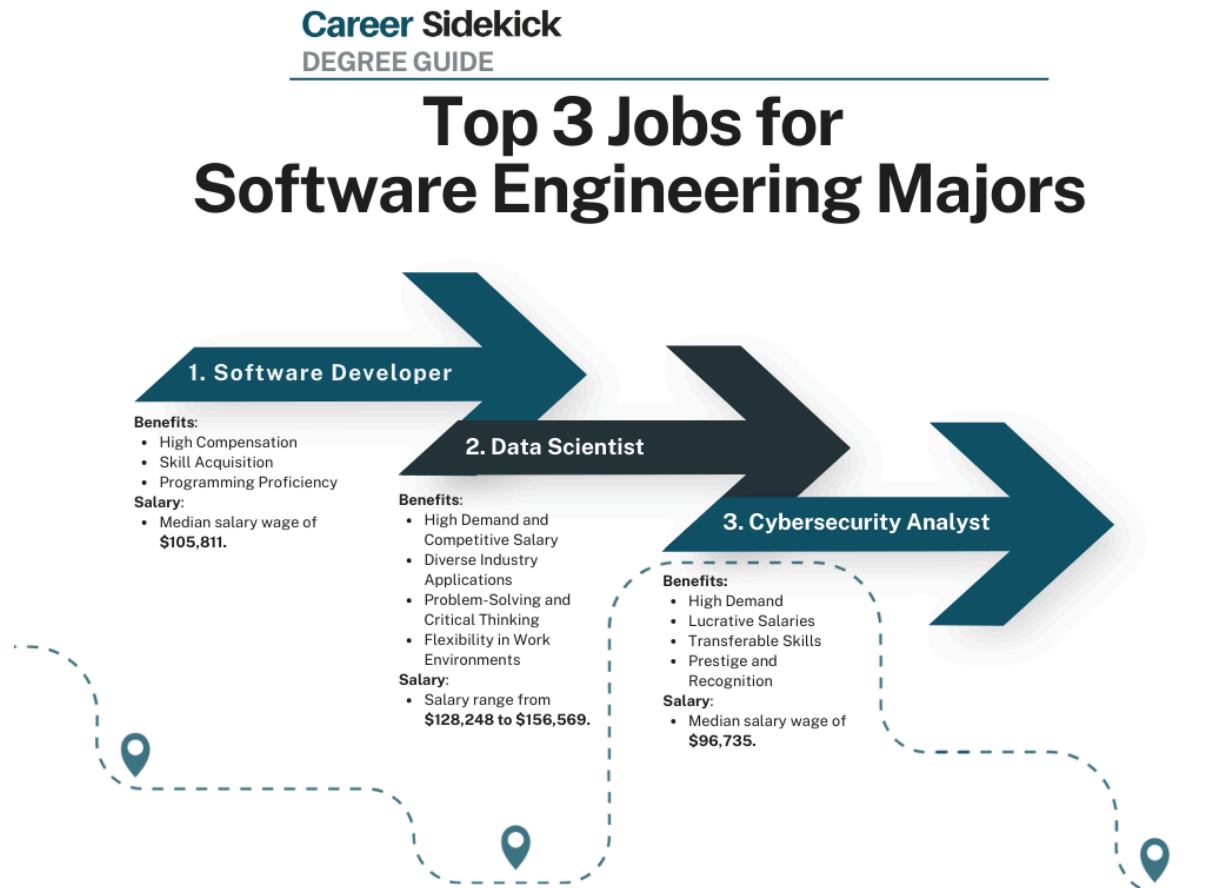


What can you do with a software engineering degree? After graduation, students can take on many different positions depending on their abilities and career aspirations. Below are some of the most common jobs in the software engineering field.



Software Developer / Programmer

Programmers are individuals who directly write code, design, and develop software, websites, mobile applications, or management systems according to the requirements of clients or businesses. They are the "heart" of the software industry, transforming ideas into tangible products that serve work and life.

A programmer's job includes: analyzing requirements, building functionality, optimizing source code, testing, and maintaining software. They typically use popular programming languages such as Java, Python, C#, PHP, JavaScript, Kotlin, Swift, etc., depending on the type of project.

This position is suitable for those with logical thinking skills, a love for creativity, patience, and a passion for technology. With good experience and abilities, programmers can advance to higher positions such as software engineer, technical team leader, or project manager.

Software Tester / QA Engineer

Software testers are responsible for testing, detecting bugs, and ensuring that software products function stably and meet requirements before being released to the market. This is a crucial position in ensuring quality assurance (QA) and the best possible user experience.

A tester's job includes: building test scripts, performing manual or automated testing, documenting bugs, collaborating with programmers to fix them, and verifying results. They typically use tools such as Selenium, JMeter, Postman, TestRail, or JIRA to manage the testing process.

This position requires logical thinking, attention to detail, carefulness, and strong analytical skills. It's also an ideal starting point for those who want to develop a career in software quality assurance or transition into a QA Leader or Business Analyst role.

System Analyst

A systems analyst acts as a bridge between the client and the engineering team, helping to translate business requirements into specific software solutions. Those in this role must have a thorough understanding of the company's business needs and the technical knowledge to advise on and design suitable systems.

The main responsibilities of a System Analyst include: gathering and analyzing user requirements, modeling operational processes, designing data structures, and collaborating with programmers to implement the product. They also participate in testing and evaluating the system's effectiveness after it is put into operation.

This position is suitable for individuals with strong communication skills, sharp analytical thinking, and a thorough understanding of both technical and business aspects. It serves as an important stepping stone to advancement to higher positions such as Project Manager or Solution Architect.

IT Project Manager

IT project management is one of the most popular choices when people ask what you can do with a software engineering degree. This person is responsible for planning, coordinating, monitoring, and ensuring the progress of IT projects, including software projects. They play a central role, connecting clients, development teams, testers, and stakeholders to ensure projects are completed on time and within budget.

The responsibilities of an IT Project Manager include: developing implementation plans, assigning tasks, managing risks, monitoring progress, resolving issues, and reporting results to management or clients. In addition to management skills, they need a thorough understanding of the software development workflow (SDLC) and technical knowledge to make accurate decisions.

This position is suitable for individuals with strategic thinking, leadership skills, strong communication abilities, and the capacity to work under high pressure. It is one of the highest-paying jobs with the greatest opportunities for advancement in the software industry.

DevOps Engineer

DevOps engineers are responsible for connecting the software development team (Developers) and the operations team (Operations), ensuring that software deployment, operation, and updates are fast, stable, and secure. This is a crucial position in businesses that use an Agile development model.

The work of a DevOps Engineer includes: automating deployment processes, managing servers and cloud infrastructure, monitoring system performance, and detecting and resolving issues promptly. They typically work with tools such as Docker, Jenkins, Kubernetes, AWS, Azure, Google Cloud, and the Git source code management system.

This position requires a strong foundation in programming, operating systems, security, and cloud computing, along with the ability to troubleshoot quickly and work effectively in a team. It's one of the "hottest" professions with high demand and attractive salaries in the software technology field.

Data Engineer

Data engineers are responsible for collecting, processing, storing, and managing data to support analysis, reporting, and decision-making within a business. They build a solid data foundation that enables data analysts and data scientists to effectively leverage information.

The work of a Data Engineer includes: designing database systems, building data processing pipelines, cleaning and transforming data from various sources, and ensuring data integrity and security. They typically use technologies such as SQL, Python, Hadoop, Spark, Kafka, Airflow, AWS, Azure, or Google Cloud.

This position is suitable for individuals with logical thinking, analytical skills, and a deep understanding of data systems. In the era of rapidly developing artificial intelligence and Big Data, data engineers are a high-demand profession with good salaries and global job opportunities.

Product Manager / Product Owner

What can you do with a software engineering degree? You can't overlook the PM and PO positions – Product Management Specialists. They are responsible for guiding, developing, and optimizing software products based on market and user needs. They act as the "conductor" in the product development process, ensuring a balance between customer desires, technical capabilities, and the company's business goals.

The responsibilities of a Product Manager (PM) or Product Owner (PO) include: market research, identifying features to develop, creating a product roadmap, and coordinating with the engineering, design, and marketing teams to bring the product to market on time. They typically work using an Agile/Scrum model, where the Product Owner represents the customer's voice within the development team.

This position is suitable for individuals with strategic thinking, market analysis skills, strong communication abilities, and technological expertise. It's a creative and leadership-oriented profession that directly impacts product success.

Maintenance & Support Engineer

Software maintenance and support specialists ensure that systems, applications, and software operate stably, securely, and efficiently after deployment. They are the "solid foundation" that helps businesses quickly resolve issues, maintain a good user experience, and minimize operational risks.

The responsibilities of a Maintenance & Support Engineer include: monitoring systems, troubleshooting technical issues, performing regular software updates, backing up data, and providing user support when problems arise. Additionally, they gather customer feedback to suggest upgrades, optimize performance, and improve features.

This position is suitable for individuals who are patient, possess strong analytical and problem-solving skills, and have a good understanding of operating systems, networks, and security. It's an essential role in any technology company, especially those with large, 24/7 operational systems.

Top Careers You Can Pursue with A Software Engineering Degree

